MARKING SCHEME

Model Paper — 04

Third Term Test

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Note (for teachers):

- Part I – A: 2 marks per question (normally 1 mark for the correct method and 1 mark for the correct final answer). Total 30 marks.
- Part I – B: 10 marks per question. Total 20 marks. Part II: candidates answer any five questions, 10 marks each. Total 50 marks. The marks for each step are given in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from a graph.

PART I – A — Answers

1. Differences: 4, 6, 8, 10, … (constant second difference 2); next differences 12, 14; 30 + 12 = 42; 42 + 14 = 56

Final answer: 42, 56

(2 marks)

2. $11001_2 = 1\times16 + 1\times8 + 0\times4 + 0\times2 + 1\times1 = 25$; $25 < 26$

Final answer: 26 is greater (26 > 25)

(2 marks)

3. $\frac{3}{5} \times \frac{5}{2} = \frac{15}{10}$

Final answer: $\frac{3}{2} = 1\frac{1}{2}$

(2 marks)

4. Increase = $\frac{20}{100} \times 250 = 50$; $250 + 50$ (or $250 \times \frac{120}{100}$)

Final answer: Rs 300

(2 marks)

5. $6ab - 9ac$; common factor $3a$

Final answer: $3a(2b - 3c)$

(2 marks)

6. $5^0 = 1$; $2^{-1} = \frac{1}{2}$; $1 + \frac{1}{2}$

Final answer: $1\frac{1}{2}$ ($= \frac{3}{2} = 1.5$)

(2 marks)

7. $(2 \times 4) \times 10^{3+2} = 8 \times 10^5$

Final answer: 8×10^5

(2 marks)

8. $3x = 9 \times 4 = 36$; $x = 36 \div 3$ (check: $\frac{3 \times 12}{4} = 9$)

Final answer: $x = 12$

(2 marks)

9. Vertically opposite angles are equal $\Rightarrow 3x = x + 40 \Rightarrow 2x = 40$ (check: $3 \times 20 = 60 = 20 + 40$)

Final answer: $x = 20$

(2 marks)

10. $x + (x + 20) + (x + 40) = 180 \Rightarrow 3x + 60 = 180 \Rightarrow 3x = 120$ (check: $40 + 60 + 80 = 180$)

Final answer: $x = 40°$

(2 marks)

11. $\pi d = 88 \Rightarrow \frac{22}{7} \times d = 88 \Rightarrow d = 88 \times \frac{7}{22}$

Final answer: 28 cm

(2 marks)

12. Pythagoras' theorem: $d^2 = 8^2 + 6^2 = 64 + 36 = 100$

Final answer: 10 cm

(2 marks)

13. $A = \{2, 4, 6, 8, 10\}$; A' = elements of U that are not in A

Final answer: {1, 3, 5, 7, 9}

(2 marks)

14. MATHEMATICS has 11 letters; the letter M occurs 2 times

Final answer: $\frac{2}{11}$

(2 marks)

15. 750 ml = 0.75 l; 1.5 + 0.75

Final answer: 2.25 litres (2.25 l)

(2 marks)

PART I – B — Answers and Mark Allocation

Question 1

(i)	
Distance for 1 l = 84 ÷ 6	**1**
= 14 km	**1**
(ii)	
Litres needed = 210 ÷ 14 (or 6 × 210 ÷ 84)	**1**
= 15 litres	**1**
(iii)	
Cost = 15 × 370	**1**
= Rs 5 550	**1**
(iv)	
Increase = $\frac{10}{100} \times 370 = 37$ (or $370 \times \frac{110}{100}$)	**1**
New price = 370 + 37 = Rs 407	**1**
(v)	
Cost = 15 × 407	**1**
= Rs 6 105 (check: 5 550 × 1.1 = 6 105)	**1**

Total marks: 10

Question 2

(i)	
Outer side = $x + 2 + 2$	**1**
$= (x + 4)$ m	**1**
(ii)	
Area = $(x + 4)^2$	**1**
$= x^2 + 8x + 16$ (m^2)	**1**
(iii)	
Area of path = area of outer square − area of lawn = $(x^2 + 8x + 16) - x^2$	**1**
$= 8x + 16$ (m^2) (shown)	**1**
(iv)	
$8x + 16 = 8 \times x + 8 \times 2$ (common factor 8)	**1**
$= 8(x + 2)$	**1**
(v)	
$8x + 16 = 96 \Rightarrow 8x = 80$	**1**
$x = 10$ (check: outer side 14 m, $14^2 - 10^2 = 196 - 100 = 96$)	**1**

Total marks: 10

PART II — Answers and Mark Allocation

Question 1

(a)	
Loss = $\frac{15}{100} \times 40\,000$	**1**
= Rs 6 000	**1**
Selling price = 40 000 − 6 000 = Rs 34 000 (or 40 000 × 0.85)	**1**
(b)	
After the first year = $2500 \times \frac{108}{100}$	**1**
= 2700	**1**
After the second year = $2700 \times \frac{95}{100}$	**1**
= 2565	**1**
(c)	
$\frac{45}{180} \times 100\%$	**2**
= 25%	**1**

Total marks: 10

Question 2

(a)	
$(x + 4)(2x - 3) = x(2x - 3) + 4(2x - 3) = 2x^2 - 3x + 8x - 12$	**1**
$= 2x^2 + 5x - 12$	**1**
(b)	
(i) $x^2 + x - 20 = (x + 5)(x - 4)$	**1**
(ii) $2x^2 - 18 = 2(x^2 - 9)$	**1**
$= 2(x - 3)(x + 3)$	**1**
(iii) $xy - 3x + 2y - 6 = x(y - 3) + 2(y - 3)$	**1**
$= (y - 3)(x + 2)$	**1**
(c)	
LCM 6a; $\frac{3}{2a} = \frac{9}{6a}$, $\frac{1}{3a} = \frac{2}{6a}$	**2**
$\frac{9}{6a} - \frac{2}{6a} = \frac{7}{6a}$	**1**

Total marks: 10

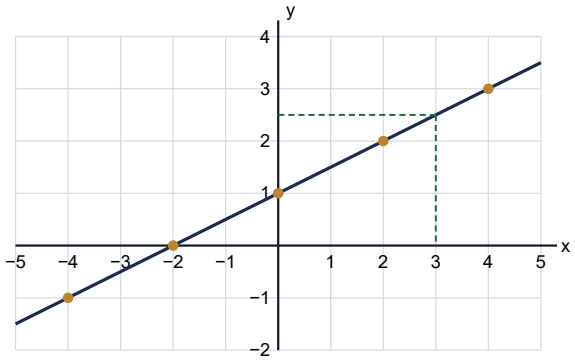
Question 3

(a)	
Son's age x, father's age 3x	**1**
After 10 years: $3x + 10 = 2(x + 10)$	**1**
$3x + 10 = 2x + 20 \Rightarrow x = 10$	**1**
Son 10 years, father 30 years (check: after 10 years $40 = 2 \times 20$)	**1**
(b)	
$2x + y = 11 \rightarrow ①$, $x - 2y = -2 \rightarrow ②$; $① \times 2$: $4x + 2y = 22$	**1**
Adding ②: $5x = 20$	**1**
$x = 4$	**1**
Substituting: $8 + y = 11 \Rightarrow y = 3$ (check: $4 - 6 = -2$)	**1**
(c)	
$60 = \frac{1}{2} \times h \times (7 + 5) = 6h$	**1**
$h = 10$	**1**

Total marks: 10

Question 4

(i)	
$x = -2 \Rightarrow y = 0$; $x = 2 \Rightarrow y = 2$ (both blanks correct)	**1**
(ii)	
Axes with a suitable scale (covering x from −4 to 4 and y from −1 to 3)	**1**
Points (−4, −1), (−2, 0), (0, 1), (2, 2), (4, 3) plotted correctly	**1**
Points joined by a straight line	**1**



Correct graph: solid line $y = \frac{1}{2}x + 1$; dashed lines show the reading for (v)

(iii)	
Gradient = $\frac{1}{2}$ (= 0.5)	**1**
Intercept = 1 (the line cuts the y-axis at (0, 1))	**1**
(iv)	
$y = 0 \Rightarrow \frac{1}{2}x + 1 = 0 \Rightarrow x = -2$; coordinates (−2, 0)	**1**
(v)	
Draw a horizontal line from y = 2.5 to the graph and read x from the x-axis	**1**
$x = 3$ (check: $\frac{1}{2} \times 3 + 1 = 2.5$) (2.9 – 3.1 accepted)	**1**
(vi)	
When $x = 6$, $y = \frac{1}{2} \times 6 + 1 = 4$, so (6, 4) lies on the line (yes) — with the reason	**1**

Total marks: 10

Question 5

(a)	
Since AB ∥ CD, $A\hat{P}Q = P\hat{Q}D$ (alternate angles) ⇒ 3x + 10 = 5x − 30 (with reason)	**2**
40 = 2x ⇒ x = 20 ($P\hat{Q}D = 5\times20 - 30 = 70° = A\hat{P}Q = 3\times20 + 10$)	**1**
$C\hat{Q}P = 180° - 70° = 110°$ (adjacent angles on the straight line CD)	**1**
(b)	
(2n − 4) × 90° = 1080° ⇒ 2n − 4 = 12	**1**
(i) n = 8 sides	**1**
(ii) One interior angle = 1080° ÷ 8 = 135° (check: exterior angle 360° ÷ 8 = 45°, 180° − 45° = 135°)	**1**
(c)	
Exterior angle = sum of the interior opposite angles ⇒ $\hat{P} + \hat{Q} = 130°$	**1**
$\hat{P} = \frac{2}{5} \times 130° = 52°$	**1**
$\hat{Q} = \frac{3}{5} \times 130° = 78°$ (check: $\hat{R} = 50°$, 52° + 78° + 50° = 180°)	**1**

Total marks: 10

Question 6

(a)(i)	
Circumference = length of wire: $2\pi r = 88 \Rightarrow 2 \times \frac{22}{7} \times r = 88$	**1**
$r = 14$ cm	**1**
Area = $\frac{22}{7} \times 14 \times 14 = 616$ cm^2	**1**
(a)(ii)	
Side of square = $88 \div 4 = 22$ cm	**1**
Area = $22 \times 22 = 484$ cm^2	**1**
(a)(iii)	
The circle encloses the larger area; $616 - 484 = 132$ cm^2 more	**1**
(b)	
Diagonal^2 = $24^2 + 7^2 = 576 + 49 = 625$	**1**
Diagonal = 25 m	**1**
Distance along the two edges = $24 + 7 = 31$ m	**1**
Distance saved = $31 - 25 = 6$ m	**1**

Total marks: 10

Question 7

(i)	
$3 + 7 + f + 4 + 2 = 20 \Rightarrow 16 + f = 20$	**1**
$f = 4$	**1**
(ii)	
Total children $= 1\times3 + 2\times7 + 3\times4 + 4\times4 + 5\times2 = 3 + 14 + 12 + 16 + 10$	**1**
$= 55$	**1**
(iii)	
Mean $= 55 \div 20$	**1**
$= 2.75$	**1**
(iv)	
Highest frequency is 7 (families with 2 children) $\Rightarrow$ mode = 2	**1**
(v)	
20 families $\Rightarrow$ median is the mean of the 10th and 11th values	**1**
Cumulative frequencies: 3, 10, 14, 18, 20 $\Rightarrow$ 10th value = 2, 11th value = 3	**1**
Median $= (2 + 3) \div 2 = 2.5$	**1**

Total marks: 10